



nextwork.org

VPC Endpoints



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Details			
Endpoint ID  vpce-09f5b141210a24e6e	Status  Available	Creation time Thursday, October 9, 2025 at 15:19:36 EDT	Endpoint type Gateway
VPC ID  vpc-075e5f2a7f88d4e5b (NextWork-vpc)	Status message -	Service name  com.amazonaws.us-east-1.s3	Private DNS names enabled No
Service region us-east-1			



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual private cloud that lets me launch AWS resources in an isolated network. It is useful because it gives control over networking, security, and access to AWS resources.

How I used Amazon VPC in this project

I used Amazon VPC to create an isolated network where I launched my EC2 instance, allowing it to securely connect and interact with my S3 bucket through a VPC endpoint, so that traffic does not go through the public internet.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was how easy it is to setup a VPC endpoint and modify the route tables.

This project took me...

This project took me 30 minutes to complete.



In the first part of my project...

Step 1 - Architecture set up

In this step, I will create a VPC, launch an EC2 Instance, and set up a VPC endpoint.

Step 2 - Connect to EC2 instance

In this step, I will connect to my EC2 instance and try to connect to the S3 bucket through the public internet.

Step 3 - Set up access keys

In this step, I will set up keys so that my EC2 has access to my AWS environment.

Step 4 - Interact with S3 bucket

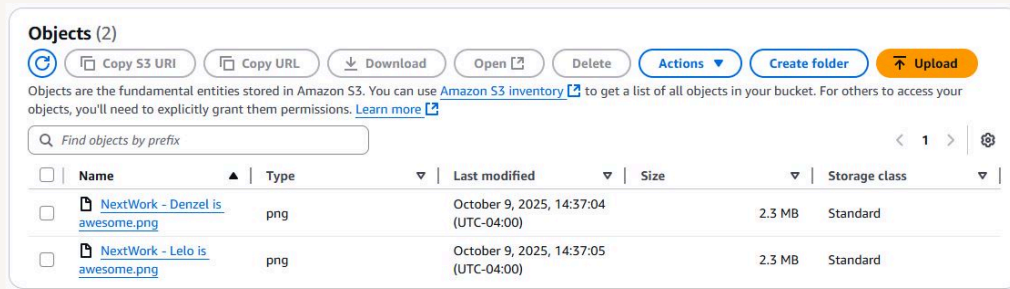
In this step, I will access my S3 bucket through my EC2 instance.



Architecture set up

I started my project by launching a VPC with a public subnet and one EC2 instance.

I also set up an S3 bucket and uploaded two objects into the bucket.





Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured the Access Key ID, Secret Access Key, Default region name, and Default output format.

Access keys are credentials made up of an access key ID and secret key that let you securely access AWS services through the CLI or SDK.

A secret access key is the private part of your access keys that works with your access key ID to authenticate your identity when using AWS CLI or SDK.

Best practice

The best practice is to use IAM roles instead of access keys since roles provide temporary credentials and are automatically managed by AWS.



Connecting to my S3 bucket

The command I ran was `aws s3 ls`. This command is used to list all the S3 buckets that are in my account.

The terminal responded, listing the bucket I had created. This indicated that the access keys I set up are working correctly because I am able to list the buckets in my account.

```
[ec2-user@ip-10-0-4-100 ~]$ aws s3 ls
2025-10-09 18:36:16 nextwork-vpc-project-mdmota
[ec2-user@ip-10-0-4-100 ~]$
```



Connecting to my S3 bucket

I also tested the command `aws s3 ls s3://nextwork-vpc-project-mdmota`, which returned a list of objects inside my bucket.

```
[ec2-user@ip-10-0-4-100 ~]$ aws s3 ls s3://nextwork-vpc-project-mdmota
2025-10-09 18:37:04      2431554 NextWork - Denzel is awesome.png
2025-10-09 18:37:05      2399812 NextWork - Lelo is awesome.png
[ec2-user@ip-10-0-4-100 ~]$
```



Uploading objects to S3

To upload a new file to my bucket, I first ran the command `sudo touch /tmp/nextwork.txt`. This command creates a new file in my EC2 instance.

The second command I ran was `aws s3 cp /tmp/nextwork.txt s3://nextwork-vpc-project-mdmota`. This command will copy the file into my S3 bucket.

The third command I ran was `aws s3 ls s3://nextwork-vpc-project-mdmota`, which validated that the file was in the S3 bucket by listing all the objects inside of the bucket.

```
[ec2-user@ip-10-0-4-100 ~]$ sudo touch /tmp/nextwork.txt
[ec2-user@ip-10-0-4-100 ~]$ aws s3 cp /tmp/nextwork.txt s3://nextwork-vpc-project-mdmota
upload: ../../tmp/nextwork.txt to s3://nextwork-vpc-project-mdmota/nextwork.txt
[ec2-user@ip-10-0-4-100 ~]$ aws s3 ls s3://nextwork-vpc-project-mdmota
2025-10-09 18:37:04      2431554 NextWork - Denzel is awesome.png
2025-10-09 18:37:05      2399812 NextWork - Lelo is awesome.png
2025-10-09 18:56:51           0 nextwork.txt
[ec2-user@ip-10-0-4-100 ~]$
```



In the second part of my project...

Step 5 - Set up a Gateway

In this step, I will create a VPC endpoint because this will allow my VPC and S3 to communicate directly rather than going over the public internet.

Step 6 - Bucket policies

In this step, I am going to create a bucket policy that will block all traffic except traffic that is coming from my VPC endpoint.

Step 7 - Update route tables

In this step, I will test the VPC endpoint setup. And fix any connectivity errors if they come up.

Step 8 - Validate endpoint connection

In this step, I will test my VPC endpoint setup. And I will restrict my VPC's access to my AWS environment.



Setting up a Gateway

A gateway is a network device that connects your VPC to the internet or other networks, allowing traffic to flow in and out.

What are endpoints?

An endpoint is a private connection that lets your VPC securely access specific AWS services without using the public internet.

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Bucket policies

A bucket policy is a set of rules that controls who can access an S3 bucket and what actions they can perform on its objects.

My bucket policy allows only resources from my VPC endpoint to access the S3 bucket, blocking all public internet traffic.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Deny",
6       "Principal": "*",
7       "Action": "s3:*",
8       "Resource": [
9         "arn:aws:s3:::nextwork-vpc-project-mdmota",
10        "arn:aws:s3:::nextwork-vpc-project-mdmota/*"
11      ],
12      "Condition": {
13        "StringNotEquals": {
14          "aws:sourceVpce": "vpce-09f5b141210a24e6e"
15        }
16      }
17    }
18  ]
19 }
20 |
```



Bucket policies

Because the bucket policy blocks access from outside the VPC endpoint, so any requests not using the endpoint get denied.

I updated the route table to direct traffic to the VPC endpoint so my VPC can reach S3 privately without going through the internet.

The screenshot displays three configuration sections in the AWS IAM console, each with a red error banner indicating a permission issue:

- Block public access (bucket settings)**: The error states, "You don't have permission to view the Block public access (bucket settings) configuration." It suggests using `s3:GetAccountPublicAccessBlock` and provides a link to "Learn more about Identity and access management in Amazon S3".
- Bucket policy**: The error states, "You don't have permission to get bucket policy." It suggests updating IAM permissions to allow `s3:GetBucketPolicy` and provides a link to "Learn more about Identity and access management in Amazon S3".
- Object Ownership**: The error states, "You don't have permission to view Object ownership (bucket settings) configuration." It suggests using `s3:GetBucketOwnershipControls` and provides a link to "Learn more about Object ownership in Amazon S3".

Each section includes an "API response" link and a "Diagnose with Amazon Q" button.



Route table updates

I added a route pointing S3 traffic to the VPC endpoint so requests stay within the VPC and don't use the internet.

After updating my public subnet's route table, my terminal could return the list of objects in my S3 bucket.

Route table: [rtb-0b6a6a37e0973a30b](#) / [NextWork-rtb-public](#) [Edit route table association](#)

Routes (3)

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-0c41ae56bbd0bfc7
pl-63a5400a	vpce-09f5b141210a24e6e



Endpoint policies

An endpoint policy is the way you can give or take away access to the VPC endpoint in a fast and granular way.

I updated my endpoint's policy by putting the Effect as Deny. I could see the effect of this right away, because when I ran the ls command in the CLI, I was not able to see the list of objects inside my S3 bucket.

```
1 {  
2   "Version": "2008-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Deny",  
6       "Principal": "*",  
7       "Action": "*",  
8       "Resource": "*"   
9     }  
10  ]  
11 }
```



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